Healthcare Data Analysis Project

Acute Malnutrition in Gaza: Analysis of the SMART 2026 Survey Report

Project Type: Healthcare and Humanitarian Data Analytics

Geographic Focus: Gaza Strip, Palestine

Primary Data Source: Gaza SMART Survey 2026 Report

Tools: Microsoft Excel, SQL, Power BI

Prepared by: Mahmoud Aref Algoul



1. Executive Summary

This project examines acute malnutrition and related child health indicators in the Gaza Strip using estimates published in the Gaza SMART Survey 2026 report.

The project aims to transform published humanitarian health statistics into structured, transparent, and decision-oriented data analysis. It focuses on understanding the prevalence of acute malnutrition, comparing estimates across geographic domains, examining confidence intervals, and communicating findings through data visualization.

The analysis includes Global Acute Malnutrition (GAM), Severe Acute Malnutrition (SAM), and combined indicators based on weight-for-height, mid-upper arm circumference, and bilateral pitting oedema, as reported in the source document.

2. Background and Problem Statement

Reliable health data are essential for understanding humanitarian needs and supporting evidence-informed decisions. In conflict-affected settings, access constraints can limit the populations reached by assessments and complicate the interpretation of health indicators.

Malnutrition is an important public health concern because it can reflect inadequate nutrition and broader vulnerabilities affecting children and their households. Interpreting malnutrition indicators requires more than comparing percentages: analysts must also consider indicator definitions, geographic coverage, confidence intervals, sampling design, and the limitations of the available data.

This project addresses the need to organize and interpret published malnutrition estimates in a clear analytical structure that can support reporting, visualization, and humanitarian health analysis.

3. Project Objectives

1. Structure published malnutrition and child-health indicators into a consistent analytical dataset.
2. Compare cGAM and cSAM estimates across the geographic domains reported in the survey.
3. Interpret confidence intervals and communicate statistical uncertainty appropriately.
4. Use Excel to organize, validate, and prepare the extracted data.
5. Use SQL to query indicators, summarize categories, and compare geographic estimates.
6. Develop Power BI dashboards to communicate key findings on healthcare, malnutrition, and childhood morbidity.
7. Identify data-quality issues and document corrections made during data preparation.
8. Explain the limitations of the published estimates and avoid conclusions that cannot be supported by the available data.

4. Data Source and Scope

The primary source is the Gaza SMART Survey 2026 report. The project extracts selected published estimates, confidence intervals, indicator definitions, and geographic comparisons from the report.

The analysis focuses on children aged 0–59 months where specified by the relevant indicators. The geographic domains include Gaza and North Gaza, Deir Al-Balah, Khan Younis and Rafah, and the Gaza Strip weighted estimate.

The source report presents survey estimates based on the population accessible to the assessment. These estimates should not automatically be interpreted as representing every person or every inaccessible area in Gaza.

The available document does not provide the child-level dataset required to independently reproduce the original weighted prevalence estimates or confidence intervals. All extracted values must therefore remain traceable to their source tables and pages.

5. Excel Data Preparation

• Organizing the Data

I used Microsoft Excel to organize the malnutrition indicators reported in the Gaza SMART Survey 2026 report.

I built a structured dataset in which each row represents an indicator for a specific geographic area. This made it easier to compare results, check values, and prepare the data for SQL and Power BI.

The dataset includes indicators related to acute malnutrition, stunting, underweight, maternal nutrition, and selected childhood illnesses.

• Dataset Structure (The Excel dataset contains the following columns)

Column	Description
indicator_id	Short, unique identifier for each indicator
indicator_name	Full name of the indicator
category	Type of health or nutrition indicator
age_group	Age group covered by the indicator
domain	Geographic area
estimate_pct	Reported prevalence (%)
ci_low	Lower confidence limit
ci_high	Upper confidence limit
page	Page number of the source report
ci_width	Difference between the upper and lower confidence limits
relative_precision	Measure used to assess uncertainty relative to the estimate
precision_flag	Project-defined precision label
who_severity	Severity interpretation, where applicable
highest_domain	Area with the highest reported estimate
lowest_domain	Area with the lowest reported estimate

- ## Data Checks

I checked the dataset for consistency before using it in further analysis.

The checks included:

- Making sure indicator IDs matched their names.
- Checking that prevalence estimates and confidence limits were entered as numbers.
- Calculating confidence interval width.
- Keeping the source page for each record.
- Checking geographic comparisons against the reported values.

For example, the Gaza Strip cSAM estimate is 0.3%, with a confidence interval from 0.1% to 0.8%. Its interval width is 0.7 percentage points.

These checks help identify inconsistencies in my extracted data. However, they do not independently verify the original survey calculations because the child-level dataset was not available.

6. SQL Analysis

- ## Counting Indicators by Category

I used the following query to count the distinct indicators in each category:

```
SELECT
    category,
    COUNT(DISTINCT indicator_id) AS total_indicators
FROM smart_indicators
GROUP BY category
ORDER BY total_indicators DESC;
```

What this query tells me: It shows how the dataset is distributed across health and nutrition categories. This helps me understand what the dataset covers before analyzing individual indicators.

- # Comparing Acute Malnutrition Indicators

I used this query to retrieve selected indicators based on weight-for-height and combined acute malnutrition classifications:

```
SELECT
    indicator_id,
    indicator_name,
    domain,
    estimate_pct,
    ci_low,
    ci_high,
    who_severity
FROM smart_indicators
WHERE indicator_id IN (
    'GAM_WHZ',
    'SAM_WHZ',
    'cGAM',
    'cSAM'
)
ORDER BY indicator_id, estimate_pct DESC;
```

What this query tells me: It returns the estimates for the selected indicators across geographic areas, together with their confidence limits. This makes it easier to compare results while keeping their uncertainty visible.

The indicators measure related but different forms of malnutrition, so I interpret each one according to its definition in the report.

• Comparing Geographic Areas

The report provides the following estimates for combined acute malnutrition:

Geographic Area	cGAM (%)	cSAM (%)
Gaza and North Gaza	1.7	0.2
Deir Al-Balah	2.0	0.5
Khan Younis and Rafah	0.7	0.2
Gaza Strip	1.3	0.3

Deir Al-Balah has the highest reported cGAM and cSAM estimates among the three geographic assessment areas. Khan Younis and Rafah have the lowest cGAM estimate.

These results show differences between the reported estimates, but they do not automatically prove that the differences are statistically significant.

The Gaza Strip estimate is reported separately as a weighted result. It should not be replaced with a simple average of the regional percentages.

• What SQL Added to the Project

Using SQL helped me:

- Retrieve the indicators relevant to the analysis.
- Summarize the dataset by category.
- Compare estimates across geographic areas.
- Keep the analysis organized and repeatable.

Because I used published report data rather than individual child records, I did not independently recalculate the survey's weighted prevalence estimates.

7. Power BI Visualization

Power BI was used to transform the structured dataset into interactive dashboards that support the exploration of healthcare indicators, acute malnutrition, and childhood morbidity in Gaza.

- ## Healthcare Overview

This page provides an overview of selected healthcare and nutrition indicators included in the project. It serves as an entry point for exploring the dataset and understanding its overall scope.

- ## Malnutrition Analysis

This page focuses on malnutrition indicators and supports comparisons across the geographic domains reported in the SMART 2026 survey. The findings must be interpreted alongside the definitions of each indicator and the associated confidence intervals.

- ## Morbidity Analysis

This page focuses on selected childhood morbidity indicators reported in the source document. It provides a way to examine reported illness estimates while recognizing that the project uses aggregated survey results rather than individual-level health records.

- ## Interpretation and Data Limitations

The Power BI dashboards present extracted and organized estimates; they do not independently reproduce the original survey calculations. Geographic differences should be interpreted cautiously, particularly where confidence intervals are wide or population access was incomplete.

The dashboards support descriptive analysis and communication of findings. They do not establish causal relationships between humanitarian conditions and health outcomes.

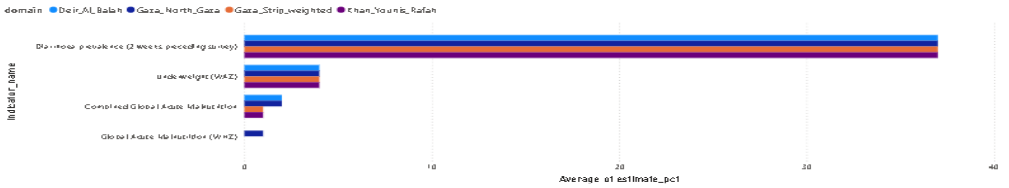
GAZA SMART SURVEY • 2026

Child Health & Nutrition Overview

Malnutrition, Morbidity & Public Health Indicators in Gaza

Diarrhoea Prevalence	Stunting Prevalence	Combined Global Acute Malnutrition	Fever Prevalence
37.00	12.00	1.00	32.00

Average of estimate_pct by indicator_name and domain



indicator_name	estimate_pct	domain	who_severity
Acute Malnutrition - non-pregnant non-lactating women (MUAC<230mm)	11.00	Deir_Al_Balah	High (≥10%)
Acute Malnutrition - pregnant/lactating women (MUAC<230mm)	11.00	Deir_Al_Balah	High (≥10%)
Diarrhoea prevalence (2 weeks preceding survey)	37.00	Deir_Al_Balah	High (≥20%) — Emergency level
Diarrhoea prevalence (2 weeks preceding survey)	37.00	Gaza_North_Gaza	High (≥20%) — Emergency level
Diarrhoea prevalence (2 weeks preceding survey)	37.00	Gaza_Strip_weighted	High (≥20%) — Emergency level
Diarrhoea prevalence (2 weeks preceding survey)	37.00	Khan_Younis_Rafah	High (≥20%) — Emergency level
Minimum Acceptable Diet	16.00	Gaza_North_Gaza	Critical (<25%)
Minimum Acceptable Diet	19.00	Deir_Al_Balah	Critical (<25%)
Minimum Acceptable Diet	22.00	Gaza_Strip_weighted	Critical (<25%)
Minimum Dietary Diversity (5+ food groups) [CORRECTED]	22.00	Deir_Al_Balah	Critical (<25%)

domain

- ☐ Deir_Al_Balah
- ☐ Gaza_North_Gaza
- ☐ Gaza_Strip_total
- ☐ Gaza_Strip_weighted
- ☐ Khan_Younis_Rafah

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Malnutrition Analysis

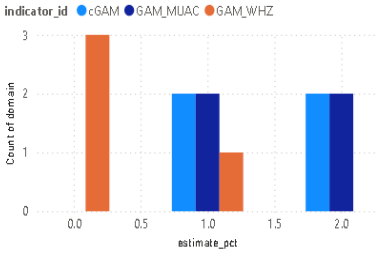
Malnutrition, Morbidity & Public Health Indicators in Gaza

indicator_name	domain	estimate_pct	ci_low	ci_high	Sum of ci_width
Combined Global Acute Malnutrition	Deir_Al_Balah	2.00	0.90	4.20	3.30
Global Acute Malnutrition (MUAC)	Deir_Al_Balah	2.00	0.80	4.30	3.50
Global Acute Malnutrition (WHZ)	Deir_Al_Balah	0.00	0.10	2.00	1.90
Combined Global Acute Malnutrition	Gaza_North_Gaza	2.00	0.80	3.60	2.80
Global Acute Malnutrition (MUAC)	Gaza_North_Gaza	2.00	0.60	3.60	3.00
Global Acute Malnutrition (WHZ)	Gaza_North_Gaza	1.00	0.30	2.00	1.70
Combined Global Acute Malnutrition	Gaza_Strip_weighted	1.00	0.80	2.20	1.40
Global Acute Malnutrition (MUAC)	Gaza_Strip_weighted	1.00	0.70	2.10	1.40
Global Acute Malnutrition (WHZ)	Gaza_Strip_weighted	0.00	0.20	1.00	0.80
Combined Global Acute Malnutrition	Khan_Younis_Rafah	1.00	0.20	2.20	2.00
Global Acute Malnutrition (MUAC)	Khan_Younis_Rafah	1.00	0.30	2.40	2.10
Total					**25.70**

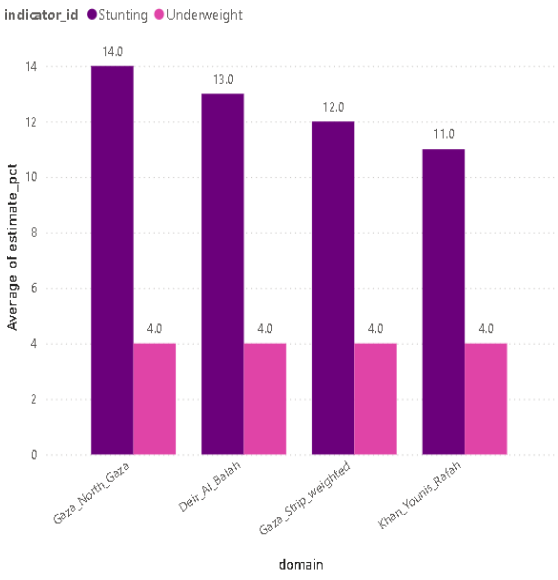
indicator_id

- ☐ cGAM
- ☐ GAM_MUAC
- ☐ GAM_WHZ

Count of domain by estimate_pct and indicator_id



Average of estimate_pct by domain and indicator_id

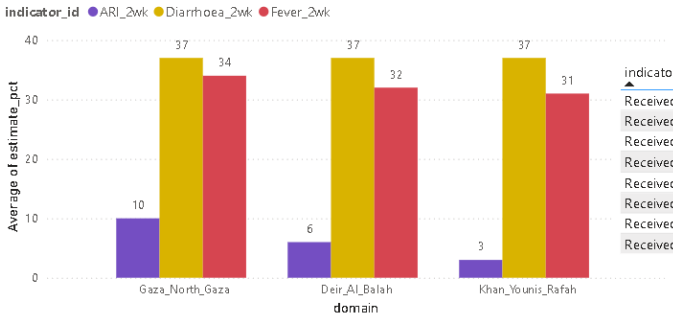


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Morbidity Analysis

Malnutrition, Morbidity & Public Health Indicators in Gaza

Average of estimate_pct by domain and indicator_id



indicator_name	domain	estimate_pct
Received both ORS and zinc (among children with diarrhoea)	Deir_Al_Balah	12.00
Received both ORS and zinc (among children with diarrhoea)	Gaza_North_Gaza	14.00
Received both ORS and zinc (among children with diarrhoea)	Gaza_Strip_weighted	12.00
Received both ORS and zinc (among children with diarrhoea)	Khan_Younis_Rafah	12.00
Received zinc (among children with diarrhoea)	Deir_Al_Balah	18.00
Received zinc (among children with diarrhoea)	Gaza_North_Gaza	24.00
Received zinc (among children with diarrhoea)	Gaza_Strip_weighted	19.00
Received zinc (among children with diarrhoea)	Khan_Younis_Rafah	15.00

8. Discussion: Interpreting Geographic Differences

1 Displacement, Aid Concentration, and Access

The geographic differences in acute malnutrition estimates should not be interpreted in isolation from the humanitarian context in which the survey was conducted. During the period covered by this assessment, population movement within the Gaza Strip was shaped by two distinct conditions:

- **Deir Al-Balah and Khan Younis/Rafah** functioned as primary displacement destinations, where large numbers of households relocated and where humanitarian aid distribution was comparatively more concentrated and accessible.
- **Gaza and North Gaza** remained under intensive military operations for extended periods, which severely restricted the entry and distribution of humanitarian assistance.

Read against this context, the lower GAM and cGAM estimates observed in Khan Younis and Rafah do not necessarily indicate a less severe underlying nutritional crisis. Rather, they may reflect **better and more consistent access to emergency food and health assistance** during the survey period. Conversely, the higher estimates recorded in Gaza and North Gaza are consistent with a population that experienced prolonged, severely constrained access to aid.

2 This Interpretation is a Hypothesis, Not a Proven Finding

This explanation is offered as a contextual interpretation, not as a statistically confirmed result. The SMART Survey did not measure aid-access rates as a direct indicator, and the relationship described above is inferred from the known humanitarian situation rather than calculated from the dataset itself.

This caution is reinforced by a further data limitation: approximately 17% of the Gaza Strip population could not be included in the survey's sampling frame due to active conflict and security constraints (Table 6, p.17). It is plausible that this excluded population was disproportionately concentrated in the areas with the most restricted access — meaning the reported estimates for Gaza and North Gaza may still understate the true severity of malnutrition in those areas, since the most affected households may not have been reachable by the assessment at all.

For this reason, this section is presented using language of likelihood ("suggests," "is consistent with," "may reflect") rather than certainty, and should be read as an interpretive discussion rather than a statistical conclusion.

3 Long-Term Cumulative Impact Versus a Single Point-in-Time Estimate

A central finding of this analysis is that acute malnutrition prevalence, while informative, captures only a narrow and recent window of nutritional status. It does not, by itself, reflect the cumulative toll of many months of prolonged crisis.

Three findings from this dataset illustrate this distinction:

1. **Chronic malnutrition (stunting) remains substantially elevated** (12.2% Gaza Strip-wide) despite acute malnutrition falling within WHO's "acceptable" threshold. Stunting develops over months of sustained inadequate nutrition and infection, and — unlike acute wasting — does not resolve quickly even if conditions subsequently improve. Its presence indicates that nutritional harm has been accumulating well before and throughout the period captured by this single survey.
2. **The ratio of stunting to combined acute malnutrition (cGAM) across domains ranges from 6.3 to 15.7**, meaning chronic, long-term damage is between six and sixteen times larger than the acute damage visible in a single snapshot. A survey conducted at any one point in time will always understate this accumulated burden, because stunting reflects the sum of conditions over many preceding months, not the conditions of the two weeks preceding data collection.
3. **Childhood morbidity indicators point to an ongoing, not episodic, burden**: diarrhoea prevalence of 37.1% in the two weeks preceding the survey is markedly higher than any acute malnutrition estimate in this dataset. Repeated infection over the course of a prolonged war degrades nutritional status continuously, even when caloric intake is nominally sufficient — a mechanism that a single malnutrition percentage cannot capture.

Taken together, these findings support a central conclusion: **a single acute malnutrition estimate, however rigorously collected, risks understating the true impact of a protracted crisis.** The low point-in-time GAM/cGAM figures reported here should be read alongside the elevated stunting prevalence and high morbidity burden — together, these indicate a population experiencing sustained, cumulative nutritional harm over the course of the war, even where the most severe acute outcomes have, so far, been contained.

4 A Treatment Gap Behind the Diarrhea Burden

Diarrhea prevalence of 37.1% is the single most urgent finding in this dataset — but the data also reveal a second, compounding problem: treatment coverage is far lower than the burden itself requires. Only 25.6% of children with diarrhea received oral rehydration salts (ORS), 18.9% received zinc, and just 12.3% received both treatments together as recommended.

This means that even when a child's illness was identified, the standard clinical response reached fewer than one in three cases. The malnutrition impact of diarrhea is therefore likely compounded by under-treatment, not caused by prevalence alone — a distinction relevant to any recommendation addressing this finding.

5 A Vitamin A Coverage Outlier That Breaks the Regional Pattern

Across most indicators in this dataset, Deir Al-Balah and Khan Younis/Rafah perform similarly, consistent with the displacement and aid-access pattern discussed in Section 8.1. Vitamin A supplementation coverage is a notable exception: Deir Al-Balah records only 46.9% coverage, sharply lower than Gaza and North Gaza (88.9%) and Khan Younis/Rafah (70.5%).

Because this indicator breaks the pattern seen elsewhere, it should not be explained by the same displacement/access hypothesis without further investigation. A supply-chain disruption specific to that domain, or a timing gap in a particular distribution round, is a more plausible explanation than general access constraints — this is flagged here as an open question rather than an interpreted finding.

6 Universal Breastfeeding Initiation Does Not Translate Into Sustained Practice

96.5% of children were ever breastfed, indicating near-universal initiation and awareness. However, this figure drops sharply at every subsequent stage of the feeding timeline: only 20.4% of infants under six months were exclusively breastfed, and dietary adequacy indicators remain low through 23 months (Minimum Dietary Diversity 30.3%; Minimum Acceptable Diet 22.4%).

This pattern suggests the underlying constraint is not maternal knowledge or willingness to breastfeed, but the conditions required to sustain exclusive and adequate feeding — maternal nutrition, stress, displacement, and access to diverse complementary foods. This distinction matters for recommendations: an intervention based on breastfeeding education would not address the actual bottleneck identified here.

7 Data Precision Varies Sharply — Severe Malnutrition Estimates Should Be Read With Caution

Not all estimates in this dataset carry equal statistical reliability. Severe Acute Malnutrition (SAM) estimates at the domain level show confidence intervals nearly nine times wider than the estimate itself (e.g., Khan Younis/Rafah SAM-WHZ: 0.2%, 95% CI 0.0–1.8%). This level of imprecision is expected given the rarity of severe cases within a sample of this size, but it means that domain-level SAM comparisons should not be used to rank areas by severity with confidence.

By contrast, Gaza Strip-wide weighted estimates — which pool a larger effective sample — carry substantially narrower intervals and support more confident interpretation. This distinction is applied consistently throughout this report: domain-level findings are described descriptively, while comparative claims rely on indicators with adequate precision.

9. Data Quality Issues Identified and Resolved

A core part of this project's analytical process involved verifying extracted data against its source before drawing conclusions from it. Two issues were identified during this verification and are documented here for transparency.

9.1 Mislabeled Indicator: Minimum Dietary Diversity

During data entry, one row labeled "Minimum Dietary Diversity (MDD)" carried the values 36.8% (Gaza and North Gaza), 40.2% (Deir Al-Balah), 50.4% (Khan Younis/Rafah), and a Gaza Strip-wide total of 43.7%.

This was flagged as inconsistent using a basic statistical check: a weighted overall estimate must fall within the range of its component domain values. Here, the reported total (43.7%) fell outside that range relative to the lowest domain value, which is mathematically impossible for a weighted average.

Returning to the source report resolved the discrepancy: the values above did not belong to the MDD indicator at all. They corresponded to a different indicator — **"Egg and/or flesh food consumption, children 6–23 months"** — which had been mislabeled during extraction. The correct MDD values, taken from Table 20 (p.36) of the source report, are:

Domain	MDD (%)	95% CI
Gaza and North Gaza	25.7	18.3–34.9
Deir Al-Balah	22.1	15.6–30.3
Khan Younis and Rafah	37.8	28.1–48.6
Gaza Strip (weighted)	30.3	25.0–36.1

Both indicators were retained in the final dataset under their correct, verified labels. This case illustrates the importance of applying basic internal-consistency checks (e.g., range validation of weighted estimates) before using extracted data for analysis, rather than assuming a labeled value is correct at face value.

- ## 2 Directional Logic Error in Automated Domain Comparison

A SQL query designed to flag whether Gaza and North Gaza represented the “best” or “worst” domain for a given indicator initially applied a single rule: the highest numeric value was labeled “worst,” and the lowest was labeled “best.”

This logic is correct only for indicators where a higher value indicates greater risk (e.g., GAM, stunting). It is incorrect for indicators where a higher value indicates a better outcome (e.g., Minimum Acceptable Diet, breastfeeding coverage). Applying the single rule uniformly produced an inverted result for Minimum Acceptable Diet: Gaza and North Gaza’s lowest MAD value (16.2%, indicating the fewest children receiving an acceptable diet) was incorrectly labeled “best.”

This was identified by manually reviewing the query output against the known direction of each indicator, and corrected by classifying indicators as either risk-coded or protective-coded before applying comparative logic. This case demonstrates that automated comparisons require an explicit encoding of each indicator’s direction; a single generic rule applied across mixed indicator types will silently produce incorrect conclusions.

10. Limitations

This analysis is based entirely on published, aggregated estimates from the Gaza SMART Survey 2026 report. It does not involve independent data collection or access to the underlying child-level dataset. The following limitations should be considered when interpreting the findings in this report.

- ## 1 No Access to Child-Level (Raw) Data

The analysis relies on estimates as published — prevalence percentages, confidence intervals, and sample sizes reported at the domain level. The individual-level survey records used to generate these estimates were not available. As a result, this project could not independently recalculate weighted prevalence values, re-run significance testing between domains, or verify the survey’s internal statistical procedures. All figures in this report are traceable to a specific page and table in the source document, but they are reproduced from published results, not derived from primary data.

2 Incomplete Population Coverage

Approximately 17% of the Gaza Strip population could not be included in the survey's sampling frame due to active conflict, military restrictions, and security-related access constraints (Table 6, p.17). The reported findings represent the accessible population only (an estimated 1.72 million people, or roughly 83% of the population) and should not be interpreted as representative of areas or households that could not be reached. As discussed in Section 8.2, this excluded population may be disproportionately concentrated in the areas facing the most severe access constraints, which could mean the true burden in those areas is understated rather than overstated.

3 Wide Confidence Intervals for Rare Outcomes

As shown in Section 8.7, estimates for low-prevalence indicators — particularly Severe Acute Malnutrition (SAM) at the domain level — carry confidence intervals that are wide relative to the estimate itself (in some cases, nearly nine times the point estimate). This is an expected statistical consequence of measuring rare events within a limited sample size, not a flaw in survey execution. However, it means that domain-level rankings for these specific indicators should be treated as indicative rather than statistically conclusive.

4 A Single Point-in-Time Survey

The SMART Survey captures a cross-sectional snapshot of the population during a defined period (31 May–15 June 2026, per the source report). It is not designed to establish trends over time or to confirm or refute findings from earlier assessments. Comparisons to prior periods referenced in this report (e.g., IPC projections) come from separate sources with different methodologies and should not be read as a continuous time series with this survey.

- ## 5 Interpretive, Not Measured, Contextual Explanations

The discussion of displacement patterns and aid access in Section 8.1–8.2 is an interpretive explanation grounded in known humanitarian context, not a variable measured directly by the survey instrument. This distinction is maintained throughout the report: statistical findings (prevalence, confidence intervals) are presented with the certainty the data supports, while contextual explanations for those findings are presented as plausible interpretations open to further verification.

11. Recommendations

- ## Strategic Implication: Why This Matters Beyond the Numbers

The central risk of this dataset, if read in isolation, is that a low acute malnutrition figure (cGAM 1.3%, within WHO's "acceptable" range) could be misinterpreted — by donors, media, or decision-makers — as evidence that the nutritional crisis in Gaza has been resolved or significantly de-escalated. This report's findings argue directly against that reading.

The geographic pattern documented in Section 8.1 shows that the low acute malnutrition figures in Deir Al-Balah and Khan Younis/Rafah are not a sign the crisis has ended — they are evidence that **humanitarian access works**: where aid could reach displaced populations consistently, the worst acute outcomes were contained. Conversely, the elevated figures in Gaza and North Gaza are not a separate or lesser problem — they are a direct, measurable consequence of restricted access during active military operations. The same population, the same war, two different outcomes, driven by one variable: whether aid could get in.

This reframes the purpose of the entire analysis. The data are not simply describing a health outcome — they are providing indirect, measurable evidence of the cost of restricted humanitarian access, using something as fundamental as children's nutritional status as the proof point.

- **1 [Headline Recommendation] Use this dataset as evidence to argue for sustained and expanded humanitarian access to Gaza and North Gaza specifically — not as a general appeal, but as a access-outcome argument.**

The data in this report make a specific, falsifiable claim: when access improved (Deir Al-Balah, Khan Younis/Rafah), acute malnutrition stayed contained. Where access remained restricted (Gaza, North Gaza), outcomes were worse. This is a stronger advocacy argument than a general humanitarian appeal, because it is grounded in a measurable before/after comparison within the same crisis and the same survey — it demonstrates, rather than asserts, that access is the single most decisive variable in this crisis.

- **2 Do not let the "acceptable" GAM classification stand alone in any summary communication.**

Every external-facing summary of this data (donor briefs, media statements, situation reports) must pair the acute malnutrition figure with the stunting figure (12.2%) and the stunting-to-cGAM ratio (up to 15.7x). Presenting GAM alone risks being used — intentionally or not — to justify reduced funding or attention, when the chronic, cumulative burden tells the opposite story.

- ## 3 Treat the 17% excluded population not as a data footnote, but as the group at greatest risk.

Because this excluded population is concentrated in the hardest-to-reach areas, their exclusion from the survey does not mean their situation is unknown or unmeasured — it is simply unmeasured by this instrument. Any response planning should assume their indicators are at least as severe as, and plausibly worse than, the worst reported domain (Gaza and North Gaza), not treat them as a neutral gap.

comparison should not be read as suggesting that displaced populations in Deir Al-Balah This or Khan Younis/Rafah were in acceptable living conditions. Displacement in these areas has typically meant relocation into overcrowded tent settlements, with limited access to clean water, sanitation, and shelter from extreme weather — conditions that are themselves strongly associated with disease transmission and nutritional risk. The diarrhoea prevalence of 37.1% recorded Gaza Strip-wide (Section 8.4), including in these same displacement areas, is consistent with this. The more favorable acute malnutrition figures in these domains therefore do not indicate a safe or adequate environment; they indicate that humanitarian food and health assistance, where it could reach people, was sufficient to prevent the most severe acute outcomes — even within a broader environment of overcrowding, poor sanitation, and high disease exposure.

This reframes the purpose of the entire analysis. The data are not simply describing a health outcome — they are providing indirect, measurable evidence of the cost of restricted humanitarian access, using something as fundamental as children's nutritional status as the proof point. Crucially, this evidence holds even though the "better-access" areas were themselves sites of harsh, disease-prone displacement conditions — tent settlements with limited sanitation — rather than places of safety or stability. This strengthens, rather than weakens, the argument: aid access made a measurable difference even against a backdrop of severe deprivation on every other dimension, which suggests that further improving access would yield even greater protective impact were baseline conditions to improve as well.